

Alpha $\leftrightarrow$ Omega Dynamics (AOD)

The Hidden Temporal Dynamics of Stokes

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Abstract

Proceeding from AFC Axiom 1 and the imported AFC Stokes identity on declared Regions, the note develops a temporal reading of the same relational continuity on declared active cuts, frames, and windows. On support read on the canonical Q4 chart, this localization yields directional $\alpha \leftrightarrow \omega$ asymmetry through backlog and reconciliation, return and enclosure, cadence, torsional tension, short-window confinement and transmutation, and native rest-channel reports in the native put/putz units and notation. The same closure and shell rules recur under window composition, yielding a fractal scale-reuse topology on the cited readout family. The visible stochastic layer remains bookkeeping on induced transitions over the deeper hidden update already carried by AFC.

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0 Introduction

0.1 Aim and stance

The local line developed below stays inside the declared AFC model universe and makes explicit the temporal reading latent in the imported Stokes consequence on declared cuts.

0.2 Upstream basis

The note depends on AFC Stokes and the AF protocol [1, 2].

0.3 Relational units and notation

The public line uses *active cut*, *frame*, *window*, *present identity*, *predecessor trace*, *successor horizon*, *net flux*, *boundary flux*, *step flux*, *window current*, *Memory path*, and *trit*. Where a unit of update is needed, the upstream notation *put* / *putz* or neutral tick / rate wording is retained. The term *cut* is reserved for the inherited relational localization object and is not reused as a unit alias.

0.4 Temporal localization (AOD reading)

On a declared readout family, the predecessor trace records the already-cut interior carried along a Memory path. The present is the active cut together with the present identity on that cut. The successor horizon is the admissible exterior continuation generated from the current Memory State, the Markov Kernel, and the declared local constraints. In this sense, temporal order is read as the emergent relational tautology of the same imported AFC identity across successive active cuts on the declared readout family.

0.5 Native cut calculus

On a declared Region $R \subseteq V$ with boundary ∂R , a temporal readout assigns ternary symbols $\tau_t \in \{-1, 0, +1\}$ on cited frames and windows in the native put/putz units and notation.

0.6 Global addendum choices

Construction order, field/readout separation, step flux versus window current, and rate normalization are fixed globally for the addendum.

0.7 Organization

After the imported relational basis and the local addendum choices, the note proceeds through local consequences, the native operator layer, a short corollary-and-hooks section, examples, and two appendices.

1 Imported relational basis

1.1 Imported Null axiom

Axiom A1 (imported Null axiom).

This addendum proceeds from AFC Axiom 1, $N \equiv \infty \equiv \chi$, by reference [1].

1.2 Imported AFC choices

The local line uses the upstream Memory State, Markov Kernel, net flux, divergence, and Region/boundary bookkeeping in the localized form used below [1].

1.3 Imported symmetry and support law

The AFC identity is symmetric at the relational level: flux is antisymmetric, internal exchanges cancel on a Region, and isotropic ignorance preserves the induced support under indistinguishability-preserving relabelling. Local asymmetry appears only after declared active cuts, knowledge profiles, and directional bookkeeping are introduced in the addendum.

1.4 Exact scope of reuse

AFC Axiom 1; AFC Choices 1–5; AFC Choice 6 / Consequence 3 where Memory-path unspooling is invoked; AFC Choice 7 / Consequence 5 where isotropic ignorance and support are invoked; AFC Corollary 1; and AFC Appendix A where stochastic bookkeeping is invoked are assumed by reference [1].

2 Local definitions and addendum choices

2.1 Memory State, active cut, and present identity

Definition 2.1 (Memory State). *The Memory State is the AFC object*

$$p : V \rightarrow [0, 1], \quad \sum_{i \in V} p(i) = 1.$$

Definition 2.2 (Active cut). *An active cut is the declared localized cut on which the imported AFC identity is read.*

Definition 2.3 (Present identity). *The present identity is the AFC Stokes identity evaluated on the active cut.*

2.2 τ readout, frame/window, and active/dwell split

Choice 2.1 (τ readout). *A declared readout map assigns a ternary symbol $\tau_t \in \{-1, 0, +1\}$ to each update on the active cut. The ternary readout is attached to the cited cut through the declared frames and windows; it does not redefine the imported cut itself.*

Definition 2.4 (Frame). *A frame f_t is the one-put readout slice on a declared active cut.*

Definition 2.5 (Window). *A window ω is a finite ordered family of frames on a fixed declared readout contract.*

Definition 2.6 (Active indicator and active/dwell split). *The active indicator is*

$$\chi_t := \mathbf{1}\{\tau_t \in \{-1, +1\}\}.$$

Active length counts the ticks with $\chi_t = 1$ and dwell length counts the complementary ticks with $\tau_t = 0$ on the same readout.

Definition 2.7 (Step flux and window current). *Step flux is the one-frame boundary-crossing quantity on a declared frame. Window current is the derived window-level accumulation of step flux over the ordered family of frames in ω .*

2.3 τ -side readout, admissibility, and loss channel

Definition 2.8 (τ readout ledger). *The declared τ -ledger records, on the cited frame/window family, the ternary symbols $\tau_t \in \{-1, 0, +1\}$, the active/dwell partition, and any declared symbol counts (N_+, N_0, N_-) . These are authoritative native ledgers.*

Definition 2.9 (Admissibility count and active set). *At tick t , the declared gating rule produces an admissible successor set A_t . The admissibility count is*

$$m_t := |A_t| \in \{1, 2, 3\}.$$

When $m_t = 1$, the realized update is fully constrained on that tick.

Definition 2.10 (Native loss channel and rest channel). *The per-tick loss burden, its accumulated total, and its windowed average are*

$$u_t^{(\text{loss})} := \log_3\left(\frac{3}{m_t}\right), \quad U^{(\text{loss})}(\omega) := \sum_{t \in \omega \cap \text{act}} u_t^{(\text{loss})},$$

$$R_{\text{rest}}(\omega) := u^{(\text{loss})}(\omega) := \frac{1}{T_{\text{eval}}} U^{(\text{loss})}(\omega) = \log_3(\Gamma_{\text{loss}}(\omega)) \quad [\text{trit/put}],$$

with

$$\Gamma_{\text{loss,raw}}(\omega) := \prod_{t \in \omega \cap \text{act}} \frac{3}{m_t}, \quad \Gamma_{\text{loss}}(\omega) := \Gamma_{\text{loss,raw}}(\omega)^{1/T_{\text{eval}}}, \quad T_{\Gamma}(\omega) := \Gamma_{\text{loss}}(\omega) - 1.$$

The native primitive is $R_{\text{rest}} = u^{(\text{loss})}$ in trit/put. A derived internal rate in putz may be reported later only after the declared attenuation/backlog step has been applied.

Definition 2.11 (Active-only versus total normalization). *Unless a row explicitly declares otherwise, reported rates are normalized by total ticks on the cited window. Active-only normalization is permitted only if T_{act} is reported alongside T_{eval} .*

2.4 Predecessor trace, successor horizon, and knowledge profile

Definition 2.12 (Predecessor trace). *The predecessor trace is the ordered family of prior active cuts and their local readouts along a Memory path.*

Definition 2.13 (Successor horizon). *The successor horizon is the admissible family of next continuations generated from the current Memory State, the Markov Kernel, and the declared local constraints.*

Choice 2.2 (Isotropic ignorance). *The AFC isotropic-ignorance posture is used unless an explicit local refinement is declared.*

Choice 2.3 (Anisotropic knowledge). *Anisotropic knowledge is a declared local refinement of distinctions or support on the active cut.*

2.5 Support and canonical Q4 chart

Choice 2.4 (Support law). *The support law used below is the AFC support law on the declared readout family.*

Choice 2.5 (Canonical Q4 chart). *At the chosen local resolution, the induced support is read on a canonical Q4 chart.*

Choice 2.6 (Axis-lift chart policy). *A declared axis-lift map associates each active ternary step with a signed toggle of exactly one of four chart axes,*

$$\pi_{\text{axis}} : (X_t, X_{t+1}) \mapsto \Delta x_t \in \{\pm e_1, \pm e_2, \pm e_3, \pm e_4\},$$

producing a Q4 chart representation of the support law used below. The chart is the declared report form used below.

Definition 2.14 (Q4 cell basis). *On the canonical chart the basic cell types are the Q4 vertices, Hamming-1 edges, square faces, cubes, and the 4-cell body. Traversal bookkeeping records per-axis signed counts on this same basis.*

2.6 Loop-witness discipline and even-cycle typing

Definition 2.15 (Loop witness). *A loop witness is a closure witness on the induced support.*

Definition 2.16 (Two-cycle return, four-cycle enclosure, and repeating closure signature). *A two-cycle return is the minimal return closure. A four-cycle enclosure is the minimal nondegenerate enclosure on the canonical chart. Repeating closure signatures are the declared even-cycle classifications built from these two witnesses.*

Definition 2.17 (Level index and beat closure). *For any cited closure witness κ with active half-length ledger $L_{\pm}(\kappa)$, define the level index*

$$\lambda(\kappa) := \frac{L_{\pm}(\kappa)}{4}.$$

The primitive beat closure is the minimal return witness with $L_{\pm} = 2$. The first nondegenerate enclosure on the canonical Q4 chart has $L_{\pm} = 4$.

2.7 Closure subclasses and minimal cage-band typing

Definition 2.18 (Closure predicate). *A closure occurrence κ fires on a segment s when the active displacement closes on the cited readout,*

$$\sum_{t \in s} \tau_t = 0,$$

and the declared detector admissibility criteria hold on that same segment.

Definition 2.19 (Beat-knot and cage-knot). *A knot is a closure occurrence treated as a localized curl on the wavelet stream. A beat-knot is the minimal closure beat with $L_{\pm} = 2$. A cage-knot is a closure occurrence that additionally encloses a 2-cell under the canonical Q_4 chart on the same declared detector policy.*

Definition 2.20 (Cage as a two-knot object and minimal cage-band). *A cage report is a two-knot object*

$$\text{Cage}(M; \omega) := (\kappa_{\text{in}}, \kappa_{\text{out}}),$$

where κ_{out} is a cage-knot and κ_{in} is a knot nested inside κ_{out} under the canonical Q_4 chart and the declared motif-to-region extractor. The minimal cage-band length signature is reported as $(L_{\text{in}} : L_{\text{out}}) = (4:6)$.

Definition 2.21 (Outer enclosure token and motif id). *The outer enclosure token η_{out} records the reported outer enclosure class on the cited window. A motif id M is a stable label for a detected class of occurrences under declared detector settings and perturbation tolerances.*

2.8 $\alpha \leftrightarrow \omega$ bookkeeping, backlog, and crowning

Definition 2.22 (α tally and ω obligation). *α denotes the local interior tally. ω denotes the closure or return obligation on the same active cut.*

Definition 2.23 (Backlog and reconciliation). *Backlog is the unresolved mismatch between the local tally and the closure obligation. Reconciliation is the declared balancing process that services that backlog.*

Definition 2.24 (Crowning). *Crowning is the reported directional closure distinction that first stabilizes a local asymmetry under the declared crown predicate.*

Definition 2.25 (Drift). *Drift is the reported phase or transport bias that persists across the declared active-cut family on a cited readout class.*

2.9 AO field, wavelet, shell/cadence interface, and saturation

Definition 2.26 (AO field). *The AO field is the ordered bundle of local AOD quantities attached to an active cut and its unspooling.*

Definition 2.27 (Wavelet). *A wavelet is a finite closure-carrying Memory-path segment equipped with ordered cutwise field quantities.*

Definition 2.28 (Shell-band extraction and cadence extractor interface). *A shell-band extractor packages the cited closure/return/phase ledger into an ordered family of reported shell bands. A cadence extractor acts on those bands to produce cycle counts, phase accumulators, rotor accumulators, and drift witnesses. No continuum frequency law is assumed.*

Definition 2.29 (Saturation). *Saturation is the occupancy of declared traversal lanes or closure seats on the local readout.*

2.10 Push/pull, effective charge, and export/radiation typing

Definition 2.30 (Push/pull boundary current). *Push/pull is the chirality-resolved boundary-current bookkeeping read from the same oriented boundary histories used for lag and torsion. The sign convention must be declared once for the cited readout family.*

Definition 2.31 (Effective-charge proxy). *With chirality-resolved boundary totals*

$$J_\alpha(M; \omega) := \sum_{t \in \omega} \sum_{(i,j) \in \partial R(M; \omega)} \phi_{ij}(t) \mathbf{1}\{\chi(t) = \alpha\},$$

$$J_\Omega(M; \omega) := \sum_{t \in \omega} \sum_{(i,j) \in \partial R(M; \omega)} \phi_{ij}(t) \mathbf{1}\{\chi(t) = \Omega\},$$

we define the effective-charge proxy by

$$q_{\text{eff}}(M; \omega) := \mathcal{N}(J_\alpha(M; \omega) - J_\Omega(M; \omega)),$$

with normalization $\mathcal{N}$ declared by the cited operator.

Definition 2.32 (Ω -phase lag). *The chirality-resolved phase-lag statistic is the declared difference between paired Ω and α phase histories on the cited readout family. Its pairing policy must be explicit.*

Definition 2.33 (Radiating/export window and emitted packet witness). *A window is radiating/exporting for motif M if the declared shell predicate fails by nonzero net export and/or elevated boundary mutual information on the same readout family. The emitted/exported packet witness is the wavelet packet crossing the declared cut on that window.*

2.11 B^* extraction, histogram support, and Jensen discipline

Definition 2.34 (Return-duration extraction). *For each declared return or closure occurrence κ , the return-duration extraction policy emits a burden sample*

$$b(\kappa) := e_\alpha(\kappa) + e_\Omega(\kappa) + f(\kappa), \quad f(\kappa) \in \mathbb{Z}_{\geq 1},$$

where e_α, e_Ω are the one-way outbound/return durations and f is the measured bridge/reflection count under the cited policy.

Definition 2.35 (Histogram, support mass, and dominant return class). *Let $\text{Occ}(M; \omega)$ denote the cited set of motif occurrences. With declared weights $w(\kappa) \geq 0$ (TTL weighting allowed but not required), define the return-duration histogram*

$$H_B(b; M, \omega) := \sum_{\kappa \in \text{Occ}(M; \omega)} w(\kappa) \mathbf{1}\{b(\kappa) = b\},$$

and the support mass on rung b

$$\Lambda_b(M; \omega) := H_B(b; M, \omega).$$

The dominant return class is

$$B^*(M; \omega) := \arg \max_b H_B(b; M, \omega),$$

with the estimator functional explicitly declared whenever ties or trimming rules are present.

Choice 2.7 (Hard-freeze pivot identity bookkeeping). *Hard-freeze is a declared limiting case used to expose the integer accounting spine. Under the minimal-pivot idealization one sets*

$$b = 2e + f, \quad f \in \mathbb{Z}_{\geq 1},$$

so that the canonical minimal case $e = 68, f = 1$ yields the dominant burden rung $b = 137$.

Remark 2.1 (Jensen discipline). *Histogram and loss aggregation must preserve the native order of operations. If $T_{\text{eval}} = n_1 + n_2 + n_3$ and*

$$\bar{m} := \frac{n_1 + 2n_2 + 3n_3}{T_{\text{eval}}},$$

then the exact loss-average on the declared window is

$$R_{\text{rest}}(\omega) = \frac{n_1 + n_2 \log_3 \left(\frac{3}{2} \right)}{T_{\text{eval}}},$$

whereas the post-averaged surrogate would be

$$\log_3 \left(\frac{3}{\bar{m}} \right).$$

These are not equal in general; equivalently,

$$J_{\text{gap}}(\omega) := \frac{n_1 + n_2 \log_3 \left(\frac{3}{2} \right)}{T_{\text{eval}}} - \log_3 \left(\frac{3}{\bar{m}} \right) \geq 0,$$

with equality only on the degenerate uniform-admissibility cases. Any rung or attenuation summary built from $b(\kappa)$ is therefore aggregated on the declared sample set before any later summary or rendering.

Choice 2.8 (Shell-band extractor). *A shell-band extractor packages the cited closure, return, and dressing stack into an ordered family of shell bands on the same window. The extractor policy id must be declared and stable across reruns.*

Definition 2.36 (Shell signature). *On a cited motif-conditioned window ω , the shell signature is the coarse fingerprint*

$$\text{Sig}(M; \omega) := (\lambda\text{-band}, B^*\text{-class}, \rho\text{-winding class}, V\text{-class}).$$

It is assembled from the level index, dominant return class, winding report, and support signature on the same declared shell-band extraction policy.

Definition 2.37 (Confinement predicate and proximity criterion). *A motif-conditioned window is declared confined when it is shell-qualified, carries the cited B^* -dominance on the same shell-signature class, and satisfies the declared proximity criterion for the relevant traverser/closure family together with the corresponding cage-band / outer-enclosure requirement on the same cited window. The transmutation trigger is the declared failure of that confinement predicate on the same cited window.*

2.12 Traverser / closure naming rule, field inventory, and structural constraints

Definition 2.38 (Traverser / closure rule). *The suffix `-on` denotes a traverser class. The suffix `-ad` denotes a closure or seat class. Traverser precedes closure in composed names.*

Definition 2.39 (Public AO field families and structural key). *The public AO field families admitted in this addendum are the two-seat family `duon/duad` and the four-seat family `tetron/tetrad`, together with declared structural subclasses and *O*-prefixed chirality tags on the same field report. Structural examples such as `tetron:1:2:6`, `tetron:1:2:8`, `tetron:1:2:10`, `tetron:3:3:6`, and `tetrad:3:4:6` are public native identifiers on cited windows.*

Definition 2.40 (Structural constraints). *Seat / traversal constraints, closure-lock constraints, and saturation policy are declared on the cited family key before any subclass persistence or transmutation claim is made.*

2.13 Observer families, coupling, chirality, polyrotor phases, torsional tension, and cadence

Definition 2.41 (Observer family and coupling). *An observer family is a declared family of Memory-State traces. Coupling is a shared narrowing relation or path convergence on common successor support.*

Definition 2.42 (Chirality). *Chirality is the persistent handedness of asymmetry under unspooling.*

Definition 2.43 (Polyrotor phase family). *A polyrotor phase family is a finite indexed family of rotor-phase readouts on the same active-cut family.*

Definition 2.44 (Torsional tension and torsional loss). *Torsional tension is the accumulated unresolved phase disagreement or reconciliation debt. Torsional loss is the admissibility lost because that tension remains unresolved.*

Definition 2.45 (Cadence). *Cadence is the declared local recurrence-rate report quantity on the cited readout class. The dominant return class B^* is defined earlier by the declared return-duration histogram and is reused here only by reference.*

2.14 Field persistence, decay, and recruitment

Definition 2.46 (TTL hazard and right-censoring). *For a declared motif id M , detector policy, and rupture predicate H , the lifetime observable is the birth–death duration*

$$\text{TTL}_{\text{put}}(M; \omega) := t_{\text{death}} - t_{\text{birth}} \in \mathbb{N},$$

with right-censoring declared when the cited window ends before rupture. TTL weighting or filtering may be used only under an explicit policy id.

Definition 2.47 (Decay, by-product census, and rupture predicate). *Decay is the typed end of persistence under the declared rupture predicate. Product and by-product census are the typed post-death motif counts on declared post-death windows. None of these alters the underlying AO field; they are report objects on the same ledgers.*

Definition 2.48 (Freezing, rebalance dwell, transmutation, and dynamic recruitment). *A cited window is freezing when sustained admissibility restriction and/or sustained CK mismatch indicates that the current representation is saturating its capacity. Rebalance events are explicit operator-logged events with nonzero dwell cost. Transmutation is a change of motif signature class across windows with explicit dwell cost. Dynamic volume recruitment is the growth of effective closure range or outer enclosure used to preserve admissibility and shell qualification under continued load.*

Definition 2.49 (Normalized speed and rest-frame window). *Using the declared axis-traversal ledgers, define the normalized speed*

$$v(\omega) := \frac{\|\Delta R(\omega)\|_2}{T_{\text{act}}(\omega)} \in [0, 1].$$

A window is rest-frame qualified when it is shell-qualified and $v(\omega) \leq \varepsilon_v$ under the declared tolerance.

3 Consequences

3.1 Consequence 1: Symmetry of the imported identity and local asymmetry on active cuts

Consequence 3.1. *The imported AFC identity is symmetric at the relational level. Local asymmetry appears only when that identity is evaluated on active cuts under a declared knowledge profile and directional $\alpha \leftrightarrow \omega$ bookkeeping. Crowning is a local closure consequence on that same active-cut family.*

Depends on. Definitions 2.1 to 2.3, 2.22 and 2.24 and choices 2.2 and 2.3.

Local substructure. Inherited cut balance; isotropic symmetry; anisotropic local readout; crowning.

Derivation sketch. From the imported AFC antisymmetry and internal-cancellation identities, together with the declared localization to active cuts and the declared knowledge profile; no external law is invoked.

Remark 3.1. *Symmetry is inherited. Asymmetry is a localized report.*

3.2 Consequence 2: Fractal scale-reuse topology on inherited support

Consequence 3.2. *The inherited support law is reused across clustering and local unspooling scales. At the chosen local resolution, the inherited support is read on a canonical Q4 chart, and the first nondegenerate enclosure on that chart is a four-cycle.*

Depends on. Choices 2.2 and 2.4 to 2.6 and definitions 2.14 and 2.16.

Local substructure. Clustering and scale reuse; inherited support law; canonical Q4 chart; minimal four-cycle enclosure.

Derivation sketch. From the imported isotropic-topology consequence together with the declared axis-lift chart policy and the Q4 cell basis. Fractal scale reuse here records the recurrence of the same support law under clustering and unspooling on the cited readout family.

3.3 Consequence 3: Two-cycle return, four-cycle enclosure, and repeating closure signatures

Consequence 3.3. *On the declared local chart, two-cycle returns and four-cycle enclosures form the minimal even-cycle closure witnesses. Repeating closure signatures are built from these witnesses and classify the local closure algebra on the cited readout.*

Depends on. Choices 2.5 and 2.6 and definitions 2.14 to 2.16.

Local substructure. Two-cycle return; four-cycle enclosure; repeating closure signatures.

Derivation sketch. A two-step backtrack closes the minimal return. The first nondegenerate enclosure on the canonical Q4 chart is a four-cycle square face. Repeating closure signatures are finite compositions of those witness types on the same inherited support.

3.4 Consequence 4: $\alpha \leftrightarrow \omega$ backlog, successor narrowing, and reconciliation

Consequence 3.4. *The local interior tally α and the closure obligation ω need not agree on an active cut. Their mismatch is reported as backlog. The successor horizon is narrowed by that backlog and by any declared coupling, and reconciliation services the resulting debt.*

Depends on. Choice 2.1 and definitions 2.6, 2.9, 2.10, 2.12, 2.13, 2.22, 2.23 and 2.41.

Local substructure. α tally; ω obligation; backlog; coupled successor narrowing; reconciliation; admissibility-loss channel.

Derivation sketch. The imported present identity is evaluated on the active cut. When the realized return obligation is not discharged on that same cut family, the unresolved debt is carried as backlog. Declared coupling and backlog policies reduce the admissible successor family, and the loss channel records the same reduction through m_t and $\nu^{(\text{loss})}$. The Markov Kernel remains the declared update rule on the cited state; backlog acts through realized clocking and admissibility on the same readout family.

3.5 Consequence 5: Wavelet passage, field saturation, and chained trit carriage

Consequence 3.5. *Wavelet passage on the inherited support carries trit bookkeeping across closure witnesses. On a declared readout, traverser/closure saturation constrains that passage and produces chained trit carriage on AFC Memory paths.*

Depends on. Choice 2.1 and definitions 2.6, 2.10, 2.26, 2.27, 2.29 and 2.38.

Local substructure. Wavelet passage; AO field; traverser/closure saturation; chained trit carriage; native loss channel.

Derivation sketch. From the declared ternary readout and active/dwell split together with the wavelet and saturation definitions on Memory paths. The visible stochastic layer remains bookkeeping over a deeper hidden nonlinear update.

3.6 Consequence 6: Coupling, chirality, polyrotor phases, and torsional tension

Consequence 3.6. *Coupled readouts can carry persistent chirality and a finite family of rotor-phase reports. Unresolved phase disagreement on such a family is reported as torsional tension and its corresponding torsional loss. The same disagreement admits a native torsional residual witness on the declared boundary current histories.*

Depends on. Definitions 2.10, 2.25, 2.41 to 2.44 and 4.9.

Local substructure. Observer-family coupling; chirality; polyrotor phase family; torsional tension; torsional loss; torsional residual.

Derivation sketch. From the declared coupling relation and the declared phase-family / torsion bookkeeping on the same active-cut family. When phase disagreement persists, the chirality-resolved boundary histories fail to cancel after alignment, giving a nonzero torsional residual and a positive loss channel on the same readout. The torsional residual is the native boundary witness used below for that disagreement.

3.7 Consequence 7: Cadence and dominant return class

Consequence 3.7. *The resolved local field carries a cadence spectrum and a dominant return class. Drift and reconciliation clocks organize the cadence report and the reported dominant rung B^* on the same readout family.*

Depends on. Definitions 2.10, 2.25, 2.28 and 2.45.

Local substructure. Cadence spectrum; drift; dominant return class; loss-rate channel.

Derivation sketch. From repeated return accounting on the active-cut family together with the declared cadence extractor, phase/rotor accumulators, and dominant-return estimator. The same readout supports both drift witnesses and the return-class histogram whose dominant rung is B^* .

3.8 Consequence 8: Confinement, short-window persistence, and transmutation thresholds

Consequence 3.8. *Under the declared shell-band extractor, shell signature, confinement predicate, and proximity criterion, a cited tetron/tetrad window may be reported as a confined short-window persistence class. Failure of the same predicate on the same shell-signature class triggers transmutation or recruitment thresholds on that window.*

Depends on. Choice 2.8, definitions 2.18 to 2.21, 2.36, 2.37, 2.39 and 2.40, and consequences 3.4, 3.6 and 3.7.

Local substructure. Shell-qualified window; beat-knot / cage-knot typing; minimal cage-band (4:6); outer enclosure token; B^* -dominance; shell-signature class; proximity criterion; confinement witness; transmutation threshold; torsional-load relation.

Derivation sketch. On the cited motif-conditioned window, the closure predicate yields beat-knot and cage-knot classes on the canonical Q4 chart. The minimal cage-band and outer enclosure token supply the enclosure-side witness used by the shell-band extractor and shell signature. Shell qualification, dominant return class, and shell-signature stability then provide the persistence side of the report, while unresolved torsional load, backlog forcing, and loss accumulation provide the failure side. The same declared proximity criterion separates sustained short-window persistence from transmutation / recruitment on that window family.

4 Native operators

This section records the compact native operator layer required for the addendum calculus. All operators below are discrete bookkeeping on declared traces, active cuts, frames/windows, and inherited support.

4.1 Q4 chart and closure operators

Definition 4.1 (Q4 chart and closure operators). *Given the axis-lift chart policy of Choice 2.6, the Q4 chart operator records signed single-axis toggles on the canonical chart, the derived level index $\lambda = L_{\pm}/4$, the primitive beat-closure witness $L_{\pm} = 2$, the beat-knot / cage-knot split, the minimal cage-band witness (4:6), and the first nondegenerate enclosure as a four-cycle witness. The chart basis consists of vertices, Hamming-1 edges, square faces, cubes, and the 4-cell body; two-cycle returns are minimal returns and four-cycle enclosures are the first nondegenerate closure class on that basis.*

Depends on. Choices 2.5 and 2.6 and definitions 2.14, 2.16 and 2.18 to 2.20.

Local substructure. Axis-lift chart policy; Q4 cell basis; beat-knot / cage-knot typing; minimal cage-band (4:6); minimal four-cycle enclosure.

Derivation sketch. From the declared chart policy and the inherited support law, active ternary steps map to single-axis toggles on Q4. The closure predicate yields beat-knot versus cage-knot classes on the same chart, the minimal two-knot cage report is typed by (4:6), and the first nondegenerate enclosure in this cubical basis is a four-cycle.

4.2 Kernel-clock reconciliation, native loss channel, and rest channel

Definition 4.2 (Kernel-clock reconciliation operator). *Let S_t be the declared local wavefront state, $A_t(S_t)$ the admissible successor set, and $P_t(\cdot \mid S_t)$ the declared local choice kernel on that set. Let β_t be the backlog witness and let F_t be the forced-dwell predicate. The realized update law is the reconciliation operator*

$$\tilde{P}_t(\tau \mid S_t, \beta_t) := \begin{cases} \delta_0(\tau), & \text{if } F_t \text{ holds,} \\ \frac{P_t(\tau \mid S_t)}{\sum_{\sigma \in A_t(S_t)} P_t(\sigma \mid S_t)}, & \text{if } \tau \in A_t(S_t) \text{ and } F_t \text{ does not hold,} \\ 0, & \text{otherwise.} \end{cases}$$

The kernel expresses local choice; backlog and forced dwell express clock-side debt service. The latter do not reparameterize the former.

Definition 4.3 (Derived putz rest rate). *With a declared attenuation accumulator $A_B(M; \omega)$ and burden reference $B_0 := 1 \text{ put}$, define the derived internal rest rate by*

$$R_{\text{rest}}^{(\text{putz})}(M; \omega) := \frac{T_\Gamma(M; \omega)}{B_0} A_B(M; \omega) \quad [\text{putz}].$$

The primitive remains $R_{\text{rest}} = u^{(\text{loss})}$ in trit/put.

Depends on. Choice 2.1 and definitions 2.6, 2.9, 2.10 and 2.23.

Local substructure. Admissibility count m_t ; forced dwell; $\Gamma_{\text{loss,raw}}$; Γ_{loss} ; T_Γ ; primitive trit/put channel; derived putz channel.

Derivation sketch. From the declared gating rule and the active/dwell split, admissibility restriction yields $m_t < 3$ and therefore positive loss on constrained ticks. Averaging on the cited active-cut family gives the native rest channel; the putz rate is the corresponding internal rate on the same ledger.

4.3 Wavelet transport, trit carriage, and saturation

Definition 4.4 (Wavelet transport operator). *A wavelet transport operator packages a finite closure-carrying Memory-path segment together with its ordered cutwise field quantities, traverser/closure saturation, and chained trit carriage on the same active-cut family.*

Definition 4.5 (Wavelet mode and transport witnesses). *For a wavelet segment e on a cited window ω , let*

$$A(e; \omega) := N_+(e; \omega) + N_-(e; \omega), \quad Z(e; \omega) := N_0(e; \omega), \quad S(e; \omega) := N_+(e; \omega) - N_-(e; \omega),$$

$$E(e; \omega) := A(e; \omega) + Z(e; \omega) = |\omega \cap e|.$$

The native ledgers are (A, Z, S, E) . Derived reporting witnesses are

$$\kappa_{\text{prog}} := \frac{A}{E}, \quad c_{\text{seat}} := \frac{Z}{E} = 1 - \kappa_{\text{prog}}, \quad \xi := \frac{S}{E}.$$

These are dictionary/rendering compressions only; the integer ledgers remain authoritative.

Definition 4.6 (Coupling exposure and propensity). *For a declared wavelet class s on a cited readout family ω , let $N_{\text{exp}}(s; \omega)$ be the exposure count and $N_{\text{cpl}}(s; \omega)$ the counted couplings attributed to those exposures. The empirical coupling propensity is*

$$p_{\text{cpl}}(s; \omega) := \frac{N_{\text{cpl}}(s; \omega)}{\max(1, N_{\text{exp}}(s; \omega))}.$$

This is the native counting contract used on the cited ledger.

Depends on. Definitions 2.10, 2.26, 2.27, 2.29 and 2.38.

Local substructure. Wavelet segment; saturation; chained trit carriage; progression and coupling-seat witnesses; exposure/coupling counts.

Derivation sketch. From the declared ternary readout and the wavelet/saturation ledgers on Memory paths. The visible stochastic layer is bookkeeping over the same cutwise transport operators.

4.4 Push/pull, effective charge, and export/radiation operators

Definition 4.7 (Push/pull boundary bookkeeping). *Push/pull is the chirality-resolved boundary-current bookkeeping on the cited readout family. With a fixed sign convention,*

$$J_{\alpha}(M; \omega) := \sum_{t \in \omega} \sum_{(i,j) \in \partial R(M; \omega)} \phi_{ij}(t) \mathbf{1}\{\chi(t) = \alpha\},$$

$$J_{\Omega}(M; \omega) := \sum_{t \in \omega} \sum_{(i,j) \in \partial R(M; \omega)} \phi_{ij}(t) \mathbf{1}\{\chi(t) = \Omega\}.$$

The same readout supports the effective-charge proxy

$$q_{\text{eff}}(M; \omega) := \mathcal{N}(J_{\alpha}(M; \omega) - J_{\Omega}(M; \omega))$$

and the phase-lag witness $\Delta \Phi_{\Omega|\alpha}(M; \omega)$ under the declared pairing policy.

Definition 4.8 (Radiating/export window and emitted packet). *A window ω is radiating/exporting for motif M when net boundary export on $\partial R(M; \omega)$ is nonzero beyond tolerance and/or the boundary mutual-information witness exceeds the declared shell bound. The emitted/exported packet witness is the wavelet packet crossing the declared cut on that window.*

Depends on. Definitions 2.30 to 2.33, 4.5 and 4.6.

Local substructure. Push/pull sign convention; chirality-resolved boundary totals; effective-charge proxy; export witness; emitted packet.

Derivation sketch. From the same oriented boundary-current histories used for lag, torsion, and shell/export classification. No external electrodynamic or thermodynamic law is invoked.

4.5 Torsional curl and torsional residual

Definition 4.9 (Torsional curl and residual). *Let $R(M; \omega)$ be the declared Region extractor for motif M on the cited active-cut family. With a fixed boundary orientation, define the chirality-resolved boundary circulation*

$$\text{Curl}_\chi(M; \omega) := \sum_{t \in \omega} \sum_{(i,j) \in \partial R(M; \omega)} \text{sgn}_{ij} J_{ij}^\chi(t), \quad \chi \in \{\alpha, \Omega\},$$

and the torsional curl difference

$$\text{Curl}^g(M; \omega) := \text{Curl}_\alpha(M; \omega) - \text{Curl}_\Omega(M; \omega).$$

The torsional residual is

$$T_{\text{tors}}(M; \omega) := \sum_{t \in \omega} \sum_{(i,j) \in \partial R(M; \omega)} \text{sgn}_{ij} (J_{ij}^\alpha(t) - J_{ij}^\Omega(t)) = \text{Curl}^g(M; \omega).$$

By linearity of the imported AFC Stokes pairing, the same quantity has the region-side form

$$\sum_{t \in \omega} \sum_{i \in R(M; \omega)} \text{div}(J^\alpha(t) - J^\Omega(t))(i) = T_{\text{tors}}(M; \omega).$$

In the symmetric zero-lag limit,

$$J^\alpha = J^\Omega \implies T_{\text{tors}}(M; \omega) = 0.$$

Depends on. Definitions 2.23, 2.32, 2.42 and 2.44.

Local substructure. Chirality-resolved boundary currents; torsional curl; boundary-side residual; region-side equivalent; zero-lag limit.

Derivation sketch. From the chirality-resolved boundary histories on the same active-cut family and the imported AFC Stokes pairing between boundary and region accounting.

Remark 4.1. *This is the native Stokes-style witness for unresolved torsional imbalance on the cited readout family.*

4.6 Cadence and polyrotor extractors

Definition 4.10 (Cadence extractor and shell-resolved accumulators). *For a declared shell band or active-cut family, the cadence extractor returns cycle counts $N_k^{\text{cyc}}(\omega)$ together with the shell-signature class and shell-resolved phase and rotor accumulators*

$$\Phi_k(\omega) := \sum_{t \in \omega} \Delta \Phi_k(t), \quad R_k(\omega) := \sum_{t \in \omega} \Delta R_k(t),$$

and drift witnesses

$$\delta \Phi_{ij}(\omega) := \Phi_i(\omega) - \Phi_j(\omega), \quad \delta R_{ij}(\omega) := R_i(\omega) - R_j(\omega).$$

The corresponding cadence witness is

$$\nu_k(\omega) := \frac{N_k^{\text{cyc}}(\omega)}{|\omega|},$$

so the reported cadence spectrum is $\mathcal{N}(M; \omega) = \{\nu_1(\omega), \dots, \nu_K(\omega)\}$.

Depends on. Definitions 2.25, 2.28, 2.43 and 2.45.

Local substructure. Cadence extractor; phase/rotor accumulators; drift witnesses; cadence spectrum; dominant return class.

Derivation sketch. From the declared shell-band extraction and phase/rotor ledgers on the same cited readout family.

Remark 4.2. *On shell-qualified windows, recurrent closure and phase-locking may be read as standing wavelet configurations on the cited support. Their unresolved phase disagreement is reported through backlog, torsional residual, cadence shift, and the native rest-channel on that same window. The dominant return class B^* records the leading return-population class of the cited window under the declared extraction policy.*

4.7 B^* ladder and histogram operators

Definition 4.11 (B^* ladder and histogram operators). *Given the return-duration extraction of Definition 2.34, the weighted histogram*

$$H_B(b; M, \omega) := \sum_{\kappa \in \text{Occ}(M; \omega)} w(\kappa) \mathbf{1}\{b(\kappa) = b\}$$

induces the support mass

$$\Lambda_b(M; \omega) := H_B(b; M; \omega)$$

and the dominant rung

$$B^*(M; \omega) := \arg \max_b H_B(b; M; \omega).$$

The hard-freeze identity

$$b = 2e + f, \quad f \in \mathbb{Z}_{\geq 1},$$

exposes the integer spine of the ladder. Under the minimal-pivot idealization $f = 1$, the canonical one-way excess $e = 68$ yields the dominant burden rung $b = 137$.

Definition 4.12 (Jensen discipline for the ladder). *Histogram and attenuation summaries must be formed on the declared sample set before downstream compression. If $T_{\text{eval}} = n_1 + n_2 + n_3$ and*

$$\bar{m} := \frac{n_1 + 2n_2 + 3n_3}{T_{\text{eval}}},$$

then the exact native loss-average is

$$\frac{n_1 + n_2 \log_3\left(\frac{3}{2}\right)}{T_{\text{eval}}},$$

not the post-averaged surrogate $\log_3(3/\bar{m})$. Equivalently, the Jensen gap

$$J_{\text{gap}}(\omega) := \frac{n_1 + n_2 \log_3\left(\frac{3}{2}\right)}{T_{\text{eval}}} - \log_3\left(\frac{3}{\bar{m}}\right) \geq 0$$

must be respected on the cited admissibility ledger. Likewise, one must not replace histogram- or attenuation-level quantities such as

$$\sum_b H_B(b; M; \omega) b^{-\hat{\Lambda}(M; \omega)}$$

by expressions built from averaged $b(\kappa)$ unless the aggregation policy explicitly licenses that reduction.

Definition 4.13 (Dominant-rung attenuation). *Using the same stack ledger $S(M; \omega)$ and burden samples $b(\kappa)$ already required to compute $B^*(M; \omega)$, define the exact return attenuation accumulator*

$$A_B^{\text{exact}}(M; \omega) := \prod_{\kappa \in S(M; \omega)} \left(\frac{b(\kappa)}{B_0} \right)^{-L_{\pm}(\kappa)} \left(\frac{B^*(M; \omega)}{B_0} \right)^{-\delta_{\text{core}}(M; \omega)}.$$

Let the stack-load exponent be

$$\hat{\Lambda}(M; \omega) := \sum_{\kappa \in S(M; \omega)} L_{\pm}(\kappa) + \delta_{\text{core}}(M; \omega).$$

Under the declared dominant-rung approximation $b(\kappa) \approx B^(M; \omega)$ on the cited stack,*

$$A_B(M; \omega) = \left(\frac{B^*(M; \omega)}{B_0} \right)^{-\hat{\Lambda}(M; \omega)}.$$

Thus Λ_b remains the histogram support mass, while $\hat{\Lambda}$ is the stack-load exponent used in attenuation.

Depends on. Definitions 2.16, 2.34 and 2.35, choice 2.7, and remark 2.1.

Local substructure. Return-duration extraction; histogram; support mass; dominant rung; shell-signature class; Jensen discipline; hard-freeze 137 spine; attenuation accumulator.

Derivation sketch. From the declared return-duration samples $b(\kappa)$ and the cited weighting/filter policy, then by algebra on the stack ledger.

4.8 Finite path sums, least reconciliation, ternary interaction gate, and diagrammatic renderings

Definition 4.14 (Finite path sum). *A finite path sum is a finite sum over declared Memory paths, with each term given by a finite product of ledger-derived factors such as admissibility counts, chirality tags, phase factors, coupling factors, and attenuation weights.*

Definition 4.15 (Least reconciliation). *Least reconciliation is the discrete minimization of a declared cost functional over admissible trace families. The cost may include loss burden, dwell cost, boundary residual, and return-debt service on the same active-cut family.*

Definition 4.16 (Ternary interaction gate). *The primitive wavelet tick alphabet is $\tau \in \{-1, 0, +1\}$. The balanced ternary gate is defined by identifying $\{-1, 0, +1\}$ with $\mathbb{Z}_3$ via $\chi(-1) = 2$, $\chi(0) = 0$, $\chi(+1) = 1$ and setting*

$$a \oplus_3 b := \chi^{-1}((\chi(a) + \chi(b)) \bmod 3).$$

This implements cancellation/cascade bookkeeping such as $+1 \oplus_3 (-1) = 0$ and $+1 \oplus_3 (+1) = -1$.

Definition 4.17 (Collision couplings and diagrammatic renderings). *Push/pull collision couplings of duon/duad and tetron/tetrad classes are represented as native bookkeeping on declared traces. Cut diagrams, closure diagrams, path-family diagrams, and finite diagrammatic renderings package the same ledgers in compact report form.*

Depends on. Definitions 2.27, 2.39, 2.41, 4.7 and 4.13.

Local substructure. Finite path sums; least reconciliation; ternary interaction gate; collision couplings; diagrammatic renderings; export packets.

Derivation sketch. From finite sums and products over declared traces on the same AO field and wavelet ledgers. Diagrammatic language packages those same ledgers in a compact report form.

Remark 4.3. *Finite path sums and diagrammatic renderings record the same cut bookkeeping on the cited window; they are report forms of the native line. They package already-declared AO field, wavelet, boundary-current, torsional, cadence, and return-ladder ledgers.*

4.9 Field persistence and lifecycle

Definition 4.18 (Field persistence and lifecycle). *For a declared motif id M and cited window family ω , the persistence operator records the birth tick, death tick, TTL hazard, censoring status, rupture type, and any product/by-product census reported after rupture. In native units,*

$$\text{TTL}_{\text{put}}(M; \omega) = t_{\text{death}} - t_{\text{birth}}.$$

When TTL is used in histograms or ladder weighting, the weighting rule must be declared on the same manifest.

Depends on. Definitions 2.39, 2.46 and 2.47.

Local substructure. TTL hazard; rupture predicate; right-censoring; decay; by-product census.

Derivation sketch. Pure integer/tick bookkeeping on the same motif-conditioned ledgers under the declared persistence contract.

4.10 Freezing, rebalance, transmutation, and dynamic recruitment

Definition 4.19 (Freezing, confinement witness, and rebalance operators). *Freezing is the reported sustained admissibility restriction on a cited motif-conditioned stream. The confinement witness is the operator-side realization of the declared confinement predicate on the same shell-qualified window and shell-signature class. Rebalance dwell is the explicit debt-service dwell attached to emission, absorption, transmutation, or recruitment events. Dynamic volume recruitment is the growth of the effective closure range when freezing persists.*

Depends on. Definitions 2.9, 2.23, 2.25, 2.36, 2.37 and 2.48.

Local substructure. Cross-shell drift; backlog witness; forced dwell; shell-signature class; B^* -anchored persistence; confinement witness; failure criterion; transmutation / recruitment of outer closure range.

Derivation sketch. On the cited readout family,

$$\text{cross-shell drift} \Rightarrow \beta_t > 0 \Rightarrow \text{forced dwell} \Rightarrow m_t < 3 \Rightarrow R_{\text{rest}} > 0.$$

When the same shell-qualified window also satisfies the declared proximity and B^* -dominance condition, the window is reported as confined. Failure of that witness triggers the declared transmutation event operator and, if sustained, dynamic recruitment of the outer closure range.

4.11 Native rest-channel derivation

Definition 4.20 (Native rest-channel derivation). *The native rest-facing chain is the declared operator composition*

$$m_t \rightarrow u_t^{(\text{loss})} \rightarrow \Gamma_{\text{loss}} \rightarrow B^* \rightarrow \hat{\Lambda} \rightarrow A_B \rightarrow R_{\text{rest}} \rightarrow R_{\text{rest}}^{(\text{putz})}.$$

Here

$$u_t^{(\text{loss})} := \log_3 \left(\frac{3}{m_t} \right), \quad U^{(\text{loss})}(M; \omega) := \sum_{t \in (\omega \cap M)_{\text{act}}} u_t^{(\text{loss})},$$

$$R_{\text{rest}}(M; \omega) = u^{(\text{loss})}(M; \omega) := \frac{1}{T_{\text{eval}}(M; \omega)} U^{(\text{loss})}(M; \omega) = \log_3(\Gamma_{\text{loss}}(M; \omega)) \quad [\text{trit/put}],$$

with

$$\Gamma_{\text{loss}}(M; \omega) := \left(\prod_{t \in (\omega \cap M)_{\text{act}}} \frac{3}{m_t} \right)^{1/T_{\text{eval}}(M; \omega)}, \quad T_{\Gamma}(M; \omega) := \Gamma_{\text{loss}}(M; \omega) - 1,$$

$$b(\kappa) = e_{\alpha}(\kappa) + e_{\Omega}(\kappa) + f(\kappa), \quad B^*(M; \omega) = \arg \max_b H_B(b; M; \omega),$$

$$\hat{\Lambda}(M; \omega) := \sum_{\kappa \in S(M; \omega)} L_{\pm}(\kappa) + \delta_{\text{core}}(M; \omega), \quad A_B(M; \omega) = \left(\frac{B^*(M; \omega)}{B_0} \right)^{-\hat{\Lambda}(M; \omega)},$$

and the derived internal rate is

$$R_{\text{rest}}^{(\text{putz})}(M; \omega) := \frac{T_{\Gamma}(M; \omega)}{B_0} A_B(M; \omega) \quad [\text{putz}].$$

Depends on. Definitions 2.9, 2.10, 2.34, 2.35 and 4.13, choice 2.7, and remark 2.1.

Local substructure. Admissibility count; native loss channel; return-duration ledger; dominant rung; stack load; attenuation accumulator; primitive trit/put channel; derived putz rate.

Derivation sketch. Direct algebra on the cited admissibility and return ledgers. The zero-lag/unconstrained limit $m_t = 3$ yields $R_{\text{rest}} = 0$ and $T_{\Gamma} = 0$.

4.12 Momentum-like native proxy and rest-frame windows

Definition 4.21 (Momentum-like native proxy). *Using the normalized speed $v(\omega)$ on a cited window, define the momentum-like native proxy by*

$$p_{\text{prox}}(M; \omega) := E_{\text{tot}}(M; \omega) v(\omega),$$

where E_{tot} is the declared total putz-valued rate assembled from the native rest-facing channel and any declared additional native totals on the same ledger. This is the native proxy channel assembled on the cited ledger.

Depends on. Definitions 2.49 and 4.20.

Local substructure. Normalized speed; rest-frame window class; native proxy channel.

Derivation sketch. From axis-traversal speed on the cited window and the already-declared native rate channels.

4.13 Radiation and export dynamics

Definition 4.22 (Radiation and export dynamics). *For a motif-conditioned Region extractor $R(M; \omega)$ with boundary $\partial R(M; \omega)$, define shell qualification by*

$$\text{shell}(M; \omega) \iff \text{net_export}(M; \omega) = 0 \text{ within tolerance} \wedge MI_{\partial}(M; \omega) \leq \varepsilon_I,$$

and define the radiating/export class by

$$\text{rad}(M; \omega) \iff \text{net_export}(M; \omega) \neq 0 \vee MI_{\partial}(M; \omega) > \varepsilon_I.$$

The emission/export packet witness is the wavelet packet whose boundary crossing certifies the export on the same cited ledgers.

Depends on. Definitions 2.27, 2.33, 4.6 and 4.7.

Local substructure. Motif→Region extractor; boundary MI export; shell-qualified window; radiating/export window; packet witness; shell falsifier.

Derivation sketch. From the same native boundary ledgers used for shell and export classification on the cited window.

4.14 Family dynamics and subclass rules

Definition 4.23 (Family dynamics and subclass rules). *The public field families carry declared dynamic constraints in addition to their naming keys: allowed occupancy states, closure-lock conditions, saturation behavior by subclass, and admissible transition rules. For duad instances, the lifetime observable is often time-to-first-interaction rather than intrinsic dissolution time; this caveat must be preserved whenever duad persistence is reported.*

Depends on. Definitions 2.29, 2.38, 2.39 and 4.18.

Local substructure. Family persistence behavior; structural key; seat/traversal constraints; closure-lock conditions; subclass saturation behavior; admissible transition rules.

Derivation sketch. Port of the native naming/lifecycle constraints into a compact operator-facing contract.

4.15 Collision and export-window calculus

Definition 4.24 (Collision and export-window calculus). *A collision window is a declared export window on which incoming wavelet ledgers, the ternary interaction gate, and any production/export tick are all recorded on the same cited trace family. On a cited short export window, the same enclosure and shell-signature classes used for confinement and transmutation also type the collision bookkeeping, so the packet witness and residual check are read on the same family-conditioned ledgers. The native residual check is*

$$\mathcal{R}_{\text{cut}} := S_{\text{out}} - (S_{\text{in}} + S_{\text{prod}}),$$

which must vanish within tolerance on the same cut ledger. Exchange rules are implemented with the balanced ternary gate $\oplus_3$ on the same declared collision/export ledger.

Depends on. Definitions 4.16, 4.17 and 4.22.

Local substructure. Collision window as export window; XOR exchange; production/export tick; packet witness; residual check.

Derivation sketch. From native tick ledgers and boundary bookkeeping on the declared collision/export window. Class-pair interaction tables, when recorded, follow the same native ledger and the same declared collision/export-window contract.

5 Corollary and falsification hooks

5.1 Corollary K1: Local AO field on active cuts

Corollary 5.1. *From Consequences 3.1 to 3.8 and the local addendum choices, one may attach a local AO field to active cuts and their unspoolings without any comparison or export layer.*

Depends on. Consequences 3.1 to 3.8 and definition 2.26.

Derivation. By composition of the local consequence chain and the native operator layer on the same active-cut family and under the same notation/choice contract.

5.2 Falsification hooks

Hook 5.1 (H1: identity symmetry / internal-cancellation audit). *A reported active cut fails if the imported identity does not exhibit the expected relational symmetry or internal-cancellation balance under the declared Region/boundary bookkeeping.*

Hook 5.2 (H2: isotropic-ignorance / anisotropic-knowledge audit). *A run fails if anisotropy claims are made without an explicit local knowledge-profile declaration or if isotropic invariance is claimed where the declared refinement breaks it.*

Hook 5.3 (H3: topology / two-cycle / four-cycle audit). *A topology report fails if the canonical Q4 chart or the two-cycle/four-cycle witness structure is not supported by the declared active-cut trace family.*

Hook 5.4 (H4: backlog / narrowing / reconciliation audit). *A backlog or reconciliation claim fails if successor narrowing, backlog, and reconciliation are not reconstructed from the same active-cut family and policy set.*

Hook 5.5 (H5: wavelet / saturation / chained-trit audit). *A wavelet-passage claim fails if the AO field, wavelet passage, saturation, and loss ledgers do not support the chained trit carriage claim on the cited Memory path.*

Hook 5.6 (H6: push/pull / chirality / torsional-residual audit). *A push/pull or torsional-tension claim fails if the chirality-resolved boundary totals, effective-charge proxy, phase-lag witness, and torsional residual are not jointly evidenced on the same readout family.*

Hook 5.7 (H7: cadence / dominant-return-class / Jensen audit). *A cadence or dominant-return-class claim fails if the declared cadence extractor and return-class estimator do not reproduce the cited cadence spectrum, B^* support mass, and the required Jensen-discipline ordering on the same readout family.*

Hook 5.8 (H8: confinement / transmutation audit). *A confinement or transmutation claim fails unless the cited window certifies the shell predicate, shell-signature class, declared B^* -dominance, declared proximity rule, and the corresponding failure threshold on the same motif-conditioned ledger.*

6 Examples

6.1 Example E1: Active cut, present identity, and symmetry/asymmetry

Depends on. Definitions 2.2, 2.3 and 2.10, choices 2.2 and 2.3, consequence 3.1, and hooks 5.1 and 5.2.

Local readout. A declared active cut c_t with one frame f_t on a finite Region R_t , together with its boundary ledger and a declared knowledge profile.

Native output. The present identity on the active cut is

$$\sum_{i \in R_t} \text{div}_t(i) = \sum_{i \in R_t, j \notin R_t} \phi_{ij}^{(t)}.$$

For a compact native witness, let $\ell_2 := \log_3\left(\frac{3}{2}\right)$ and report the same active-cut family under two declared knowledge profiles:

Row	profile	T_{eval} [put]	$N_m = (n_1, n_2, n_3)$	R_{rest} [trit/put]	$R_{\text{rest}}^{(\text{putz})}$ [putz]
E1.A	isotropic ignorance	8	(0, 2, 6)	$\ell_2/4$	$(3^{\ell_2/4} - 1)$
E1.B	anisotropic knowledge	8	(1, 2, 5)	$(1 + 2\ell_2)/8$	$(3^{(1+2\ell_2)/8} - 1)$

Here $N_m = (n_1, n_2, n_3)$ records the counts of ticks with admissibility cardinality $m_t = 1, 2, 3$ on the same active-cut family. The second row differs only by the declared local refinement and its induced admissibility restriction. The primitive is the trit/put channel; the putz column is shown only as a derived internal rate.

Hook/evidence. Hooks 5.1 and 5.2. The same cut family must support both the identity evaluation and the declared knowledge-profile tag.

6.2 Example E2: Two-cycle / four-cycle closure, backlog, and the B^* ladder

Depends on. Definitions 2.9, 2.10, 2.16, 2.22, 2.23, 2.34, 2.35 and 4.13, choice 2.7, consequences 3.3 and 3.4, and hooks 5.3, 5.4 and 5.7.

Local readout. A declared active-cut family with one two-cycle return witness, one four-cycle enclosure witness, one finite window ω , and a return ledger on the same trace family.

Native output. Let $\ell_2 := \log_3\left(\frac{3}{2}\right)$.

Row	witness	T_{eval} [put]	N_m	$b(\kappa)$ samples	B^*/Λ_b	R_{rest} [trit/put]
E2.A	two-cycle return	6	(0, 0, 6)	(2, 2, 2)	2/3	0
E2.B	four-cycle enclosure	8	(2, 2, 4)	(5, 5, 5, 7, 7)	5/3	$(2 + 2\ell_2)/8$
E2.C	hard-freeze pivot	137	(1, 0, 136)	(137); $e_\alpha = e_\Omega = 68$, $f = 1$	137/1	1/137

Row	Derived internal rate $R_{\text{rest}}^{(\text{putz})}$ [putz]	Note
E2.A	0	unconstrained baseline
E2.B	$(3^{(2+2\ell_2)/8} - 1)5^{-4}$	dominant-rung attenuation by $B^* = 5$
E2.C	$(3^{1/137} - 1)137^{-6}$	hard-freeze 137 rung witness

Histogram witness. On the same cited readout family, the rung histogram is

$$H_B(2) = 3 \text{ for E2.A, } H_B(5) = 3, H_B(7) = 2 \text{ for E2.B, } H_B(137) = 1 \text{ for E2.C.}$$

Any downstream compression must preserve the declared sample-set order; aggregating after averaging violates the Jensen discipline of Remark 2.1.

Hook/evidence. Hooks 5.3, 5.4 and 5.7. Topology, reconciliation, and ladder hooks must be satisfied on the same trace family. The cited row family records closure witness, backlog, dominant rung, attenuation, and the native rest channel on one declared readout.

6.3 Example E3: Wavelet passage, push/pull polarization, torsional residual, and cadence

Depends on. Definitions 2.26, 2.27, 2.29, 2.31, 2.33, 2.39, 2.43, 4.7 and 4.9 to 4.11, consequences 3.5 to 3.7, and hooks 5.5 to 5.7.

Local readout. A declared wavelet passage on an active-cut family with one chirality split, one polyrotor phase family, one finite window ω , one cadence extractor, and one collision/export ledger.

Native output. Let $\ell_2 := \log_3(3/2)$.

Row	field key	push/pull / q_{eff}	$N^{\text{cyc}}/ \omega $ [cyc/put]	T_{tors}	B^* / export	R_{rest} [trit/put]
E3.A	O-duon:2:1:7	balanced, $q_{\text{eff}} = 0$	1/4	0	2 / non-export	0
E3.B	duad:2:2:7	pull-biased, $q_{\text{eff}} = -1/2$	3/8	2	5 / export	$(1 + 2\ell_2)/8$
E3.C	O-tetrad:3:4:6	push-biased, $q_{\text{eff}} = +1/2$	5/12	3	137 / export	$(2 + 2\ell_2)/12$

Row	Derived internal rate $R_{\text{rest}}^{(\text{putz})}$ [putz]	Note
E3.A	0	zero-torsion baseline
E3.B	$(3^{(1+2\ell_2)/8} - 1)5^{-4}$	nonzero push/pull export row
E3.C	$(3^{(2+2\ell_2)/12} - 1)137^{-6}$	tetrad 137-rung witness

This row set makes the native chain explicit: field family, push/pull polarization, effective-charge proxy, cadence witness, torsional residual, dominant rung, export witness, primitive trit/put rest channel, and derived putz rate.

Hook/evidence. Hooks 5.5 to 5.7. The same readout family must support the wavelet, push/pull, torsional, cadence, and ladder claims. Radiation/export is witnessed on the same readout family by the declared export window and packet ledger.

A Stochastic bookkeeping notes

This appendix contains the probability deep dive and representation-level statistical material used by the same declared native calculus.

A.1 Markov bookkeeping vs hidden nonlinear update

The visible Markov description remains the stochastic representation of the deeper hidden nonlinear update on the same declared distinction family, exactly as in the imported AFC path consequence. CK mismatch on a projected stream is read as a representation-adequacy witness on that same basis.

A.2 Return and backlog probabilities

Return/backlog probabilities are admissible only as declared statistics on a cited readout family. They do not replace the integer ledgers β_t , m_t , or $b(\kappa)$, and they do not define the future; they summarize the declared successor horizon under the current Memory State and the declared policy set.

A.3 Narrowing adequacy under coupled readouts

Any empirical narrowing statistic must cite the coupling policy, the active-cut family, and the estimator functional used. Coupled-horizon narrowing is the local AOD report carried by the cited active-cut family.

A.4 Closure and crowning probabilities

Crowning and closure probabilities may be reported as declared race or hitting-time statistics on the same active-cut family. Such probabilities certify detector adequacy and regime labels on the cited readout family.

A.5 B^* histogram statistics and Jensen discipline

The B^* ladder uses a return-duration histogram H_B with declared weights and support mass. Probability-level summaries of that histogram must preserve the native sample-set order. In particular, one must not replace the histogram or attenuation operators by summaries built from

averaged burdens or averaged admissibility counts unless the aggregation policy explicitly licenses that reduction. All deeper statistical exploration of histogram tails, confidence bands, and rung mixtures belongs here rather than in the main body.

B Naming and terminology contract

This appendix records the public naming and terminology layer used by the addendum. It states the naming and serialization contract carried by the same native ledgers.

B.1 Primary body terms

Primary body terms are: active cut, frame, window, present identity, predecessor trace, successor horizon, AO field, wavelet, step flux, window current, chirality, push/pull, effective charge proxy, torsional tension, torsional loss, cadence, dominant return class, radiating/export window, and emitted/exported packet witness.

B.2 AO field and reported family naming

The public field name in this addendum is *AO field*. When a field instance is reported together with its wavelet class, the field name remains AO field and the class name is attached separately through the naming rules below.

B.3 Traverser / closure naming rule

The suffix **-on** denotes a traverser class. The suffix **-ad** denotes a closure or seat class. Traverser precedes closure in composed names.

This yields the public family/subclass pattern:

- *duon* / *duad*
- *tetron* / *tetrad*

B.4 Public family inventory

The public families admitted in this addendum are:

- **duon**: public two-seat traverser family.
- **duad**: canonical closed-seat duon subclass.
- **tetron**: public four-seat traverser family.
- **tetrad**: canonical locked tetron subclass.

B.5 WaveletSpec structural key

When an explicit serialization string is required, the canonical ASCII structural key is

$$[O-]\langle\text{type}\rangle : \langle\text{loop}\rangle : \langle\text{occ}\rangle : \langle\text{Lout}\rangle$$

with optional named properties appended after a property separator. In prose:

- `type` is one of `duon`, `duad`, `tetron`, `tetrad`;
- `loop` is the loop index on the declared chart;
- `occ` is the seat occupancy on the cited readout;
- `Lout` is the outer reported length as a decimal integer.

B.6 Chirality prefix rule

The unprefix structural key denotes an α -field instance. The prefixed form `0-type:...` denotes an Ω -field instance. Dwell/traffic is never serialized as a third chirality value.

B.7 Properties namespace

Named properties may be appended after the structural key as a report namespace. Typical properties include reported values such as q_{eff} , push/pull polarity, chirality tag, polyphase or cadence summaries, radiation/export qualifiers, shell indices, and attenuation support.

B.8 Public examples

Examples of public tags admitted by this addendum are:

- `duon:2:1:7`
- `0-duon:2:1:7`
- `duad:2:2:7`
- `0-duad:2:2:7`
- `tetron:1:2:6`
- `tetron:1:2:8`
- `tetron:1:2:10`
- `tetron:3:3:6`
- `tetrad:3:4:6`
- `0-tetron:3:3:6`
- `0-tetrad:3:4:6`

If named properties are attached, they remain attached to the cited native report row under the same naming contract.

B.9 AO field families and structural subclasses

Two-seat family. The two-seat traverser family is recorded by the public names *duon* and *duad*. The open-seat member is written `duon:2:1:<Lout>[::props]`; the closed-seat member is written `duad:2:2:<Lout>[::props]`. Effective-charge, push/pull, polyphase, cadence, and radiation/export properties attach only through the named property namespace.

Four-seat family. The four-seat traverser family is recorded by the public names *tetron* and *tetrad*. The compact addendum preserves the following structural variants in the naming contract: `tetron:1:2:6`, `tetron:1:2:8`, `tetron:1:2:10`, `tetron:3:3:6`, and `tetrad:3:4:6`. These structural variants are recorded in the public naming contract and are used by the cited native rows and operators below.

Confinement and isotope labels. Confinement names the native short-window persistence class defined on the cited tetron/tetrad ledgers. Isotope labels, when reported later, name stability subclasses on those same cited ledgers.

B.10 Push / pull, chirality, and polyphase property namespace

When properties are attached to a structural key, the recommended namespace includes reported quantities such as `qeff`, push/pull polarity, chirality tag, polyphase summaries, cadence witnesses, and radiation/export qualifiers. O-prefixed field tags (for example `O-duon`, `O-tetron`, `O-tetrad`) are part of this public report contract.

B.11 Shell, ladder, and index syntax

Only the shell and ladder syntax needed by the examples and the native operator reports is admitted here. This includes shell indices, rung support mass, and attenuation exponents where cited. The larger manual lineage may record further serialization layers on the same reported families.

B.12 Optional overlays

Dion and *tetraon* are optional overlay aliases available under the same naming contract.

B.13 Local AOD report objects

In this addendum, *frame* and *window* are declared local AOD report objects. *Step flux* is a one-frame quantity and *window current* is the derived window-level trace term used in the cited native rows.

References

- [1] Reginald Poyau. *Axiomatic Fundamentalism Calculus (AFC): The Hidden Form of Stokes*. Version V5.0, 2025. DOI: 10.5281/zenodo.17807779.

- [2] P. Reginald. *Axiomatic Fundamentalism (AF): A Logical Protocol for Traceable Research*. 2025.